



Fun Fact

There are 8 times as many atoms in a teaspoonful of water as there are teaspoonfuls of water in the Atlantic ocean.



WoW movie delayed

A World of Warcraft movie was in the making but seems like Universal has decided to delay it to 2016 <http://dgit.in/WarcraftMov>

From the labs

Creating impactful technology that actually matters

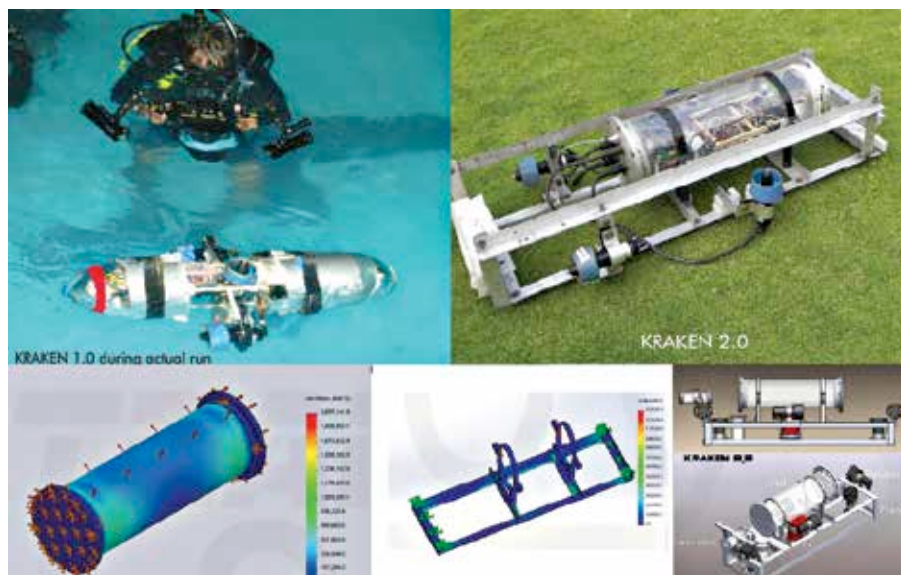
There's a lot cooking at West Bengal's premiere technical institute, as IIT Kharagpur's faculty-student research wing takes a crack at solving some of the biggest problems plaguing our time.

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The Indian Institute of Technology needs no introduction for we're all aware of their notoriously difficult entrance exams and how they're the pinnacle of technical education in the country. With such a concentration of India's best young minds, it's only apt that they take on challenges head on and find support for their cause among both faculty and alumni. In the pages that follow, we take a close look at some of the interesting research projects currently underway at IIT Kharagpur and understand how these initiatives (in various stages of their incubation) can help alleviate some of India's (and the world's) problems with the help of new applications of technology.

IIT KGP students refine AUV model, Kraken 3.0, to enter for RoboSub 2015

LOVE for BOTS – that is what has brought together a group of brilliant students from various undergraduate disciplines at IIT Kharagpur and formed “Team AUV”. They have designed and built Autonomous Underwater Vehicles (AUVs) for research and competition purposes. The project is running under the Centre for Robotics, IIT Kharagpur and is being mentored by Prof. C.S. Kumar (Mechanical Eng. Dept.), Prof. Vishwanath Nagarajan (Ocean Engineering and Naval Architecture Dept.) and Prof. Vikranth Racherla (Mechanical Eng. Dept.). Team AUV is working towards participating in the International AUVSI RoboSub Competition. This annual competition, held at San Diego, California, is a platform for



Some of the creative contraptions on display at IIT Kharagpur

industries and students to display their skills in underwater robotics.

“We live in an age where we are trying to reduce as much human effort as possible, whether that's a good thing or a bad thing, is for you to decide but until you decide we'll try our best to make your lives as laid back as possible. Our lives are affected by Bots to an extent which is beyond our imaginations. From painting our cars to cleaning our houses they do everything without complaining. So we at Kharagpur Robosoccer Students' Group (KRSSG) thought of taking off their stress by making them play soccer.”, says Abhay Kumar, (Fourth Year Undergraduate Student, Department of Ocean Engineering & Naval Architecture, Team Lead, Indian Institute of Technology, Kharagpur).

An autonomous underwater vehicle (AUV) is a robot which traverses and completes an assigned task /mission under the water surface. The team is developing

the whole vehicle, including the control systems, hardware, sensor interfacing, manufacturing etc. that is building a platform by means of which the task is to be accomplished. Secondly the team needs to encode the mission/task efficiently in the vehicle so that it can effectively exploit the resources which the vehicle is offering and thus complete the given task. The applications of such a device are numerous and can forever change the way we look at the sea.

The team is yet again representing India by taking the Kraken 3.0 to the 18th AUVSI RoboSub. The team had previously cleared qualifying rounds when competing against over 30 international teams and has since worked really hard on developing this new version, working in the Hydrodynamics lab.

Cost effective dialysis

The kidney is essential to maintaining